

Cube Values

Worksheet 7A.02.03

Name _____

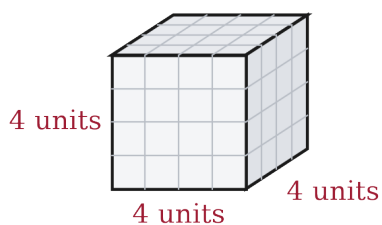
Date _____

Period _____

1. Complete the table. Each volume is the side length multiplied by itself three times.

Side s	1	2	3	4	5
Volume s^3					
Side $-s$	-1	-2	-3	-4	-5
Volume $(-s)^3$					

2. Find the volume of the cube shown.



3. A cube-shaped box has a volume of 125 cubic inches. How long is one edge?

For questions 4 through 7, find the value.

4. $\sqrt[3]{27}$

5. $\sqrt[3]{-64}$

6. $\sqrt[3]{125}$

7. $\sqrt[3]{-1}$

For questions 8 through 10, find every real solution of the equation.

8. $x^3 = 8$

9. $y^3 = -125$

10. $n^3 = 0$

11. Select ALL of the values below that make $x^3 = 64$ true.

☐ 4

☐ -4

☐ $\sqrt[3]{64}$

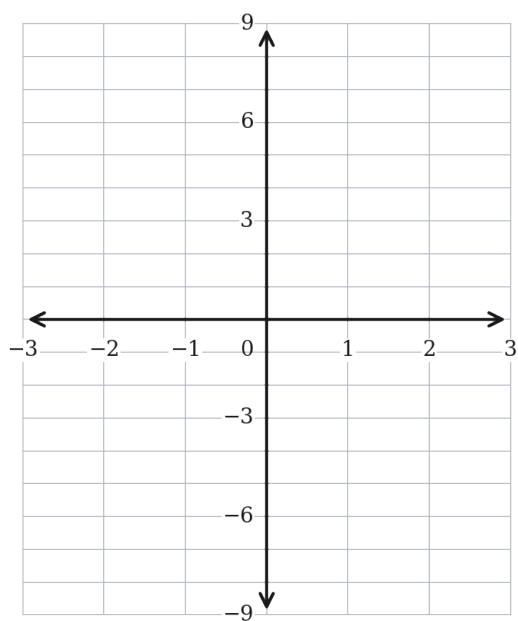
☐ $-\sqrt[3]{64}$

☐ 8

☐ $21\frac{1}{3}$

12. Complete the table of values for $y = x^3$. Then plot the five points on the grid.

x	Substitute into $y = x^3$	Coordinates (x, y)
-2		
-1		
0		
1		
2		



13. The equation $x^2 = 64$ has two solutions, but the equation $x^3 = 64$ has only one. Explain why.

STRETCH. The two questions below use cubes larger than the Grade 8 FAST will ask for. The method is identical — only the arithmetic is bigger.

14. $\sqrt[3]{216}$

15. $m^3 = 343$